

Derbyshire Shared Care Pathology Guidelines

Hyperkalaemia

Purpose of Guideline

Dealing with **adult** patients with Hyperkalaemia in the community

Definition

Serum potassium normal range is 3.5 – 5.3 mmol/L

- Potassium 5.5 – 5.9 mmol/L Mild hyperkalaemia
- Potassium 6.0 – 6.4 mmol/L Moderate hyperkalaemia
- Potassium ≥ 6.5 mmol/L Severe hyperkalaemia

Importance of Hyperkalaemia

Severe hyperkalaemia should be treated as a medical emergency and carries the risk of a cardiac event (see below)

These limits are for guidance only; the severity of clinical effects depends not only on the potassium level but also on whether the rise is acute or chronic. In many patients if the patient has acute kidney injury they will need admission to hospital. In mild to moderate hyperkalaemia, if previous results have been similar there may be no need for urgent admission.



When the potassium is ≥ 6.5 mmol/L the result will be telephoned to Derbyshire Health United when the GP practice is closed.

When is hyperkalaemia suspected?

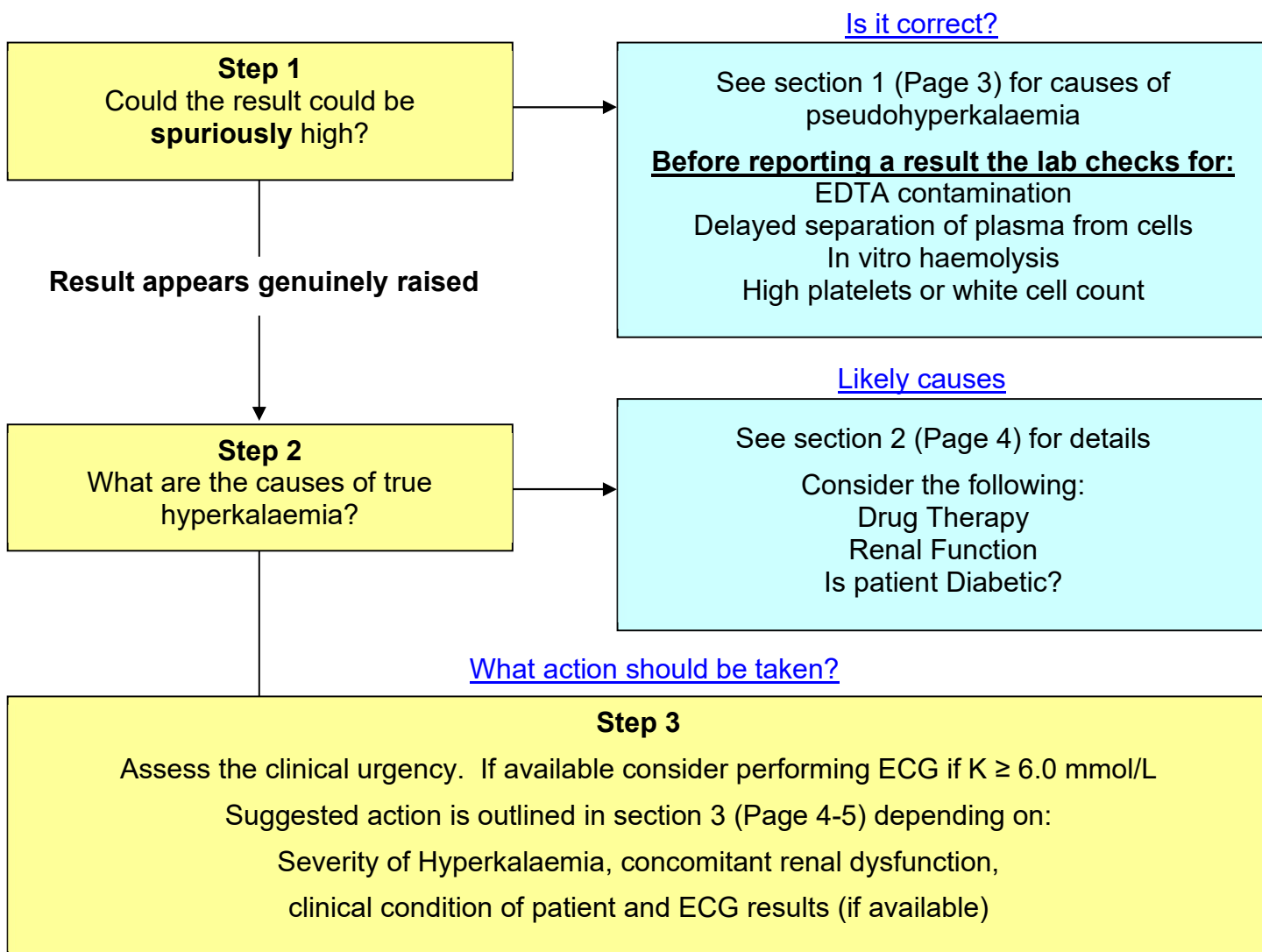
Most hyperkalaemia is an incidental finding when blood tests are done for another reason or as part of routine drug monitoring. Excessive dietary intake of potassium is not normally associated with hyperkalaemia if renal function is normal as the kidneys readily excrete any excess. **In General Practice, hyperkalaemia is most often associated with certain drugs and renal dysfunction.**⁶⁰

Clinical signs and symptoms of hyperkalaemia include:

- Arrhythmias
- Muscle weakness, fatigue, paraesthesia
- ECG changes (peaked T waves, widening of QRS complex)

However, mild/moderate hyperkalaemia is often asymptomatic. Further appropriate investigations and management (see below) should still be followed even in the absence of symptoms.

Hyperkalaemia on a blood test, What happens next?



1. Is it correct?

Pseudohyperkalaemia (artefactually increased *in vitro* potassium levels while the *in vivo* concentration is normal) is the most common reason for an isolated raised potassium result and the following causes should be considered, especially if there is no clinical evidence of hyperkalaemia.

- K-EDTA contamination of the sample from an FBC tube.
 - When using the Vacutainer system **ALWAYS** take the U&E (Gold Top) sample first. A small amount of contamination from the FBC Tube (Lavender Top) will lead to an artefactually raised potassium result.
- Delayed sample separation.
 - Potassium starts to rise in un-separated samples after 6 hours storage at room temperature (15° C - 25° C), and is usually unsuitable for analysis >12 hours after venepuncture if not separated from the cells. To enable the laboratory to spot this **ALWAYS** provide the time and date a sample is taken.
- Storage at cold temperatures.
 - At low temperatures (< 8° C) the ATPase Na-K pump becomes inactive, and there is a rapid leak of potassium out of the red blood cells. Therefore **NEVER** refrigerate samples. Depending how samples are transported and stored this same effect can be seen on very cold days.
- Thrombocytosis
 - Leakage of potassium from platelets during clotting and centrifugation can lead to falsely raised potassium results in patients with thrombocytosis. Please use a Lithium Heparin sample (Green Top) if known.
- Severe leucocytosis / Fragile white cells
 - This would be a rare cause in GP samples, but when suspected suggest telephone Duty Biochemist.
- Haemolysis *in vitro*
 - Difficult sample collection and inappropriate sample handling (e.g. shaking tube) can lead to haemolysis *in vitro*. (e.g. difficult specimen collection)

Before reporting a result the lab will usually check for the possibility of contamination with EDTA, delayed separation of plasma from cells, *in vitro* haemolysis, high platelets or white cell count and will deal with results and give advice accordingly.

2. What are the likely causes in this patient?

What drugs is the patient on?

The following drugs may contribute to hyperkalaemia:

- ACE inhibitors
- Angiotensin-II receptor antagonists ('sartans')
- Entresto (**Sacubitril with valsartan**)
- Potassium supplements
- Potassium sparing diuretics (spironolactone, amiloride)
- Finerenone
- NSAIDs
- Trimethoprim
- Heparin
- Beta-Blockers

Where possible when requesting potassium (especially in the context of repeated hyperkalaemia) please indicate if the patient is on any of the above medication, as this information is vitally useful for Pathology and Derbyshire Health United when considering if further urgent action is needed for that patient.

What is the renal function?

In chronic kidney disease, high potassium is not usually seen until late stage 4 or 5, but it may be much higher for the same level of creatinine in acute kidney injury (AKI).

Does the patient have diabetic nephropathy?

Potassium is often much higher in longstanding diabetes with apparently moderate renal impairment.

3. What action should be taken?

If available consider performing ECG if $K \geq 6.0$ mmol/L

The clinical urgency should be assessed based on

- Presence of signs and symptoms associated with hyperkalaemia
- ECG changes
- Severity of hyperkalaemia
- Evidence of renal dysfunction, i.e: acute kidney injury or chronic kidney disease

The following advice makes the presumption that the hyperkalaemia is **NOT** spurious.

MILD: Potassium 5.5 – 5.9 mmol/L

If patient is clinically well and there is no evidence of AKI, suggest recheck potassium and serum bicarbonate within 1 to 2 weeks

If the result is unexpected, suggest repeat within 3 days

Review medications

If patient is on NSAIDs or trimethoprim, stop the medications.

If patient is on ACEi/ARB/spironolactone for cardiac or renal reasons, and if it is an isolated hyperkalaemia with no AKI or change of renal function, to advise patient on low potassium diet and recheck potassium and bicarbonate in 1 week.

If persistent mild hyperkalaemia despite low potassium diet advice, or serum bicarbonate < 20 mmol/L to liaise with cardiology/renal consultant for further advice.

In general these patients do not require emergency admission

If Patient is clinically unwell or in the presence of AKI: Consider hospital referral guided by clinical circumstance and risk of further deterioration. Additional advice can be sought from renal or medical registrar/consultant (see contact details below).

MODERATE: Potassium 6.0 – 6.4 mmol/L

- Any potassium result ≥ 6.0 mmol/L should be rechecked as soon as possible unless it has been previously similar
- Drugs which can raise potassium (Page 4) should be omitted until it has been rechecked.
- Consider emergency admission to hospital. Base your decision on:
 - Clinical state of patient
 - Presence of Arrhythmias
 - Severe muscle weakness, paralysis, fatigue, paraesthesia
 - If available, ECG changes (peaked T waves, widening of QRS complex)
 - The presence of AKI
 - Rapid increase in potassium (>0.5 mmol/L within 1-2 weeks)

Refer patients to Same Day Emergency Care (SDEC) via bed bureau (UHDB) or EMU (CRH) in the first instance. For patients with an AKI or CKD 4-5, please discuss with the renal on-call.

DO NOT refer to the Emergency Department unless the patient is acutely unwell.

SEVERE Potassium ≥ 6.5 mmol/L

- Patients with potassium ≥ 6.5 mmol/L usually require urgent admission to hospital

Refer patients either to SDEC (UHDB) or EMU (CRH) or for patients with an AKI or CKD 4-5, please discuss with the renal on-call (see contacts below).

DO NOT refer to the Emergency Department unless the patient is acutely unwell.

We have seen genuine confirmed potassium results from primary care as high as 9.5 mmol/L, so do not discount very high levels as being clinically implausible.

4. Monitoring

Severity of Hyperkalaemia	Clinically well (no AKI)	Unexpected result	Clinically unwell or AKI
MILD	Repeat within 14 days	Repeat within 3 days	Consider hospital referral if indicated
K 5.5-5.9 mmol/L	Assess for cause (Drugs, diet) and address in the community		
MODERATE	Repeat within 1 working day	Repeat within 24 hours	Refer to hospital
K 6.0 - 6.4 mmol/L	Assess for cause (Drugs, diet) and address in the community or hospital		
SEVERE	Refer to hospital for immediate assessment and treatment		
K ≥ 6.5 mmol/L	Assess for cause and address during hospital admission		

Table 1: Interval for repeat blood monitoring following an episode of hyperkalaemia¹

References

1. Clinical Practice Guidelines – Treatment of Acute Hyperkalaemia in adults. UK Renal association, 2023
[UKKA Clinical Practice Guideline – Management of Hyperkalaemia in Adults – October 2023](#)

Contacts

Derby

Duty Biochemist 01332 789383 (8am to 7pm, Monday-Friday)
On Call Consultant Biochemist Via RDH switchboard, 01332 340131 (24/7)
Renal Registrar 9am to 10pm, via RDH switchboard bleep 8121
Bed Bureau 01332 789099
MAU and SDEC 01332 788707 **OR**
MAU Nurse in Charge 07917 650751

QHB

Duty Biochemist 01283 593313 (8.30 am -4pm Mon-Fri), diverts to RDH
duty biochemist out of these hours
On call Consultant Biochemist Via QHB Switchboard 01283 566333
SDEC Via QHB Switchboard 01283 566333

Chesterfield

Duty Biochemist 01246 512212 (9am to 5pm, Mon – Fri)
On Call Consultant Biochemist Via CRH switchboard, 01246 277271
A&E, EMU and CDU Via CRH switchboard

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Dr D Becker, Dr P Blackwell, Mrs H Seddon	Nov 2013	30 th Nov 2015
Dr D Becker, Dr P Blackwell, Mrs H Seddon	Dec 2015	31 st Dec 2017
Dr E Mullaney, Dr P Blackwell, Mrs H Seddon	Jan 2018	31 st Jan 2020
Dr Z Pitman, Dr K Ng, Dr E McDonald, Dr P Blackwell, Dr D Oskeira, Ms S Knowles	Dec 2024	31 st Dec 2026